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Method for cooling a heating apparatus

Abstract

A method of cooling a heating apparatus includes a heating chamber. The method includes accumulating condensate in the heating chamber, monitoring a pressure in the heating chamber, and controlling the pressure in the heating chamber by venting the heating chamber to thereby controllably permit vaporisation of the condensate that cools the heating chamber. The pressure is controlled to maintain a rate of cooling of the heating chamber within a predetermined range. A heating installation is configured to perform the method. A cooling unit may be connected to a heating apparatus to cool the heating apparatus using the method. A method may be used for retrofitting the cooling unit to a heating apparatus to form the heating installation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Stage Application under 35 U.S.C. § 371 of International Application No. PCT/EP2020/068429, filed Jun. 30, 2020, which claims the benefit of priority to United Kingdom Patent Application No. GB 1909697.3, filed Jul. 5, 2019, and the present application claims priority to and the benefit of the filing date of both of these prior applications, which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

(2) The present invention relates to a method of cooling a heating apparatus, a heating installation configured to perform the method, a cooling unit for connecting to a heating apparatus, for cooling the heating apparatus, and a method of retrofitting a cooling unit to a heating apparatus.

BACKGROUND

(3) A heating apparatus having a heating chamber may be used for industrial cooking. In such a heating apparatus, the heating chamber may be heated and foodstuffs within the heating chamber may be cooked, or walls surrounding the heating chamber may be heated and foodstuffs may be cooked on the heated walls. In such heating operations, the heating apparatus may require cleaning after each heating operation. This requires the heating apparatus to be allowed to cool, so that it can be cleaned, ready for the next heating operation.

BRIEF SUMMARY

(4) According to a first aspect, there is provided a method of cooling a heating apparatus comprising a heating chamber, the method comprising: accumulating condensate in the heating chamber; monitoring a pressure in the heating chamber; controlling the pressure in the heating chamber by venting the heating chamber to thereby controllably permit vaporisation of the condensate that cools the heating chamber; wherein the pressure is controlled to maintain a rate of cooling of the heating chamber within a predetermined range.

(5) Monitoring the pressure in the heating chamber may be done indirectly such as by monitoring pressure downstream of the heating chamber, or monitoring temperature, or force on valve etc.

(6) Vapour may be vented in a first phase until the pressure in the heating chamber is reduced to a threshold pressure. The method may comprise activating a pressure reducer in response to reaching the threshold pressure, to reduce the pressure in the heating chamber below the threshold pressure in a second phase, in order to further control the vaporisation of the condensate.

(7) The threshold pressure may be ambient pressure (i.e. atmospheric pressure). Venting in the first phase may be to ambient air (atmospheric air). The pressure reducer may be a vacuum pump or a thermocompressor, for example.

(8) Monitoring the pressure may comprise manually reviewing a pressure display. Monitoring the pressure may comprise a controller receiving a pressure parameter relating to the pressure in the heating chamber from a sensor.

(9) Accumulating the condensate may comprise receiving vapour from a vapour supply, condensing the vapour in a heat exchanger and supplying the condensate to the heating chamber.

(10) The vapour may be steam from a steam supply. The vapour supply may be a remote steam supply in that it is delivered to the heating apparatus from a steam generator remote from the heating apparatus. The remote vapour generator may supply vapour to a plurality of apparatuses, including like heating apparatuses and/or one or more other apparatuses that use vapour. The vapour supply may be a local vapour supply which is local to the heating apparatus and comprises a vapour generator. The vapour supply may be the same vapour supply which is used for heating

the heating apparatus. The vapour supply may be dedicated to the heating apparatus.

(11) The heating apparatus may be coupled to a vapour supply. The method may comprise isolating the vapour supply from the heating chamber before accumulating the condensate.

(12) The heating apparatus may be coupled to a condensate drain via a condensate line for draining condensate from the heating chamber. The method may comprise isolating the heating chamber from the condensate drain before accumulating the condensate.

(13) Supplying the condensate to the heating chamber may comprise pumping the condensate through the condensate line to the heating chamber.

(14) The heating chamber may be vented through a vent valve. The pressure reducer may be a thermocompressor which is activated by opening a motive valve connecting the vapour supply to a motive inlet of the thermocompressor. The heating chamber may be connected to a vapour inlet of the thermocompressor via the vent valve.

(15) The heating apparatus may be rotated while cooling.

(16) A predetermined amount of condensate may be accumulated in the heating chamber before venting of the heating chamber, wherein no further condensate is accumulated after the venting begins until cooling operation is completed. Alternatively, the condensate may be accumulated in the heating chamber simultaneously or alternately with venting of the heating chamber.

(17) The pressure of the heating chamber may be automatically monitored by a controller. The venting of the heating chamber may be automatically controlled with the controller based on the monitored pressure, so that cooling of the heating apparatus follows a predetermined cooling profile.

(18) The method may comprise monitoring a rate of cooling, and controlling the pressure based on the monitored rate of cooling, such that the cooling rate stays within a predetermined range. The rate of cooling may be the rate of cooling of the vapour, the rate of cooling of the condensate and/or the rate of cooling of the heating apparatus, such as a wall of the heating chamber.

(19) The method may comprise using the same vapour supply as was used in a previous heating operation to heat the heating apparatus.

(20) According to a second aspect, there is provided a heating installation configured to perform the method in accordance with the first aspect, the heating installation comprising: a heating chamber; a condensate supply configured to supply condensate to the heating chamber and a condensate line fluidically connecting the heating chamber with the condensate supply; a vent valve, and a venting line connecting the heating chamber to the vent valve, which are configured to permit venting of the heating chamber; and a pressure sensor configured to generate a pressure parameter relating to the pressure in the heating chamber.

(21) The heating installation may further comprise a pressure reducer configured to reduce pressure in the heating chamber below a threshold pressure when activated. The pressure reducer may be a thermocompressor activatable by opening a motive valve to thereby connect the vapour supply to a motive inlet of the thermocompressor. The heating chamber may be connected to a vapour inlet of the thermocompressor via the vent valve.

(22) The condensate supply may comprise a vapour supply and a heat exchanger configured to condense vapour from the vapour supply to produce condensate.

(23) The heating installation may comprise a vapour line connecting the vapour supply with the heating chamber for supplying vapour to heat the heating chamber in a heating operation. The vapour line may comprise a supply valve configured to isolate the vapour supply from the heating chamber.

(24) The heating installation may comprise a condensate drain coupled to the heating chamber via the condensate line for draining condensate from the heating chamber in the heating operation. The condensate line may comprise a condensate valve configured to isolate the condensate drain from the heating chamber.

(25) The heating installation may comprise a controller configured to perform the method in

accordance with the first aspect.

(26) According to a third aspect, there is provided a cooling unit for connecting to a heating apparatus having a heating chamber, for cooling the heating apparatus using a method in accordance with the first aspect, the cooling unit comprising: a venting line configured to be connected to the heating chamber, the venting line having a vent valve for permitting venting of the heating chamber; a condensate line configured to be connected to the heating chamber and having a condenser configured to supply condensate to the heating chamber via the condensate line.

(27) The cooling unit may comprise a pressure reducer connected to the vent valve via the venting line and configured to reduce the pressure in the heating chamber.

(28) The cooling unit may comprise a vapour generator for supplying vapour to the inlet of the condenser, which is configured to condense the vapour to supply condensate.

(29) The cooling unit may be configured to be connectable to and detachable from a vapour supply which is configured to supply vapour to the heating apparatus, so as to supply vapour to the condenser for supplying condensate when connected.

(30) The condensate line may comprise a condensate drain (steam trap) between an outlet of the condenser and the heating chamber.

(31) The condensate line of the cooling unit may be configured to be connectable to a condensate line of the heating apparatus.

(32) The pressure reducer may comprise a thermocompressor having a motive inlet which is connected to a vapour line. The vapour line may have a motive valve and may be connectable to the vapour supply for supplying motive vapour to the thermocompressor.

(33) A sensor may be configured to generate a pressure parameter relating to the pressure in the heating chamber. The heating apparatus may have a sensor which is capable of generating a pressure parameter relating to pressure within the heating chamber, and the cooling unit may be configured to receive the pressure parameter from the sensor in the heating apparatus.

(34) The cooling unit may comprise a controller configured to monitor the pressure in the heating chamber. The controller may be configured to automatically control opening of the vent valve so that the cooling of the heating apparatus follows a predetermined cooling profile.

(35) The controller may be configured to monitor the pressure in the heating chamber and to automatically control activation of the pressure reducer so that the cooling of the heating apparatus follows a predetermined cooling profile. The controller may be configured to receive a pressure parameter from a sensor in the cooling unit, or the controller may be configured to receive a pressure parameter from a sensor in the heating apparatus to monitor the pressure.

(36) Alternatively, the pressure may be monitored manually by a display showing the pressure, and the opening of the vent valve and activation of the pressure reducer may be performed manually.

(37) According to a fourth aspect, there is provided a method of retrofitting a cooling unit in accordance with the third aspect to a heating apparatus to provide a heating installation in accordance with the second aspect, the method comprising connecting the condensate line of the cooling unit and the venting line of the cooling unit with the heating chamber of the heating apparatus.

Description

DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the invention will now be described, by way of example only, with reference to the accompanying drawings, in which:

(2) FIG. 1 schematically shows a cross sectional view of a heating apparatus;

(3) FIG. 2 schematically shows a cross-sectional view of a heating installation comprising the heating apparatus of FIG. 1 and a cooling unit;

- (4) FIG. 3 is a flow chart of a method of cooling the heating installation; and
(5) FIG. 4 is a flow chart of a method of controlling venting of a heating chamber.

DETAILED DESCRIPTION

- (6) FIG. 1 shows a cross sectional view of a heating apparatus **10** comprising a heating chamber **14** in the form of a drum and defining an enclosed cavity **12**. In this example, an external wall of the drum **14** is cylindrical and extends along a longitudinal axis **15**. In other examples, the external wall of the heating chamber may define any shape.
- (7) The drum **14** is mounted on a pair of mounts **16** at respective longitudinal ends of the drum **14**. The drum **14** is rotatably mounted on the mounts **16** such that the drum can be rotated about the longitudinal axis **15** with respect to the mounts **16**. In other examples, the drum may be fixedly mounted to the mounts or any suitable support structure.
- (8) A vapour line **18** extends from outside the drum **14** through a first mount **16** and into the drum **14**, and thereby fluidically connects the enclosed cavity **12** with a vapour supply **20**. In this example, the vapour supply **20** is a steam supply. In other examples, the vapour supply may supply any suitable vapour, such as a refrigerant.
- (9) The vapour line **18** comprises a supply valve **22** configured to isolate the heating chamber **12** from the steam supply **20** when in a closed position, and to permit flow of steam from the steam supply **20** to the heating chamber **12** when in an open position.
- (10) A condensate line **24** extends from inside the drum **14** through the first mount **16** and fluidically connects the enclosed cavity **12** with a condensate drain **26**. In this example, the condensate drain **26** is a steam trap which is configured to allow condensate to drain through to a condensate sink or to be recirculated in a feedwater or condensate collection system, but prevents through-flow of steam. An end of the condensate line **24** within the drum **14** extends downwards towards the bottom of the heating chamber **12** in the manner of a dip tube, so that it can remove condensate from the bottom of the heating chamber **12**. The condensate line **24** comprises a condensate valve **28** between the heating chamber **12** and the condensate drain **26**. The condensate valve **28** is configured to isolate the enclosed cavity **12** from the condensate drain **26** when in the closed position, and to permit the flow of condensate from the enclosed cavity **12** to the condensate drain **26** when in the open position.
- (11) In use, the heating apparatus **10** is heated in a heating operation, typically to cook foodstuffs. In an example heating operation, the supply valve **22** is opened to permit flow of steam from the steam supply **20** into the heating chamber **14**, and the condensate valve **28** is opened to permit the removal of condensate from within the heating chamber **14**. The steam within the heating chamber imparts heat to the heating chamber **14** (i.e. to the wall of the heating chamber), and therefore condenses. The condensate collects at the bottom of the heating chamber **12** and is removed through the condensate drain **26** via the condensate line **24**.
- (12) The wall of the drum **14** is therefore heated so that it can be used to cook foodstuffs on an external surface of the drum **14**. For example, the wall of the drum **14** may be heated to approximately 200° C. (degrees Celsius). In this example, at the end of the heating operation, the drum **14** is cooled to approximately 50° C. (degrees Celsius) for cleaning, so that it is ready for a subsequent heating operation.
- (13) Although it has been described that the heating apparatus comprises a vapour supply for heating the heating chamber, and a condensate drain, in other examples the heating apparatus may comprise a heating chamber such as a drum which may be heated by any other means, such that the heating apparatus does not include a vapour line to a vapour supply and a condensate line to a condensate drain. For example, suitable heating means may be conductive heating elements (e.g. electric heaters), gas burners or embedded circuits for circulating a heating fluid within the walls of the chamber (rather than its cavity).
- (14) FIG. 2 shows a heating installation **100** comprising the heating apparatus **10** described above with respect to FIG. 1 connected with a cooling unit **50**. In this example, the cooling unit **50** is

retrofittable to the heating apparatus **10** (i.e. detachable from and attachable to the heating apparatus **10**) to form the heating installation **100**. In this example, the heating apparatus **10** is provided with a pressure sensor **102** which is connected to the heating chamber **12** and which is configured to generate a pressure parameter relating to the pressure in the heating chamber **12**. In some examples, the sensor may be any sensor which is configured to generate a pressure parameter relating to the pressure in the heating chamber such as a temperature sensor or a force sensor on a valve connected downstream of the heating chamber. As will be appreciated pressure is a function of temperature when a fluid is at saturation, and force across an orifice of a suitably configured valve may be a function of pressure.

(15) The heating apparatus **10** is provided with a temperature sensor **104** which is connected to the heating chamber **12** and which is configured to generate a temperature parameter related to the temperature of the heating chamber **12** or its cavity. In some examples, the sensor may be any sensor which is configured to generate a temperature parameter relating to the temperature of the heating chamber, such as a strain sensor on a wall of the heating chamber, or a pressure sensor.

(16) In some examples, the pressure sensor and temperature sensor are part of the cooling unit and are coupled to the heating chamber **14** upon installation of the cooling unit **50**.

(17) The cooling unit **50** comprises a condensate supply **52** in the form a heat exchanger **54** having an inlet fluidically connected with the steam supply **20** via a vapour supply valve **56**, and configured to condense the steam to provide condensate. An outlet of the heat exchanger **54** is connected to the condensate line **24** between the condensate valve **28** and the heating chamber **12**, via a steam trap **58**. The steam trap **58** permits through-flow of condensate from the heat exchanger **54** to the heating chamber **12**, and prevents through-flow of any steam from the heat exchanger **54** to the heating chamber **12**. In this example, the condensate supply **52** also comprises a pump **72** which is configured to pump the condensate from the heat exchanger **54** through the steam trap, and into the heating chamber **12**.

(18) In some examples, the condensate supply may be coupled to a vapour supply of any vapour, which is condensed in the heat exchanger. In other examples, the condensate supply may comprise a liquid supply (i.e. without local condensation), such as a water supply, such that a heat exchanger and steam trap are not be needed. However, if the condensate supply is a liquid water supply, the heating installation may require a blowdown apparatus in order to remove impurities that may otherwise build up in the heating chamber.

(19) Condensing vapour from a vapour supply **20** to supply condensate ensures that the condensate is clean (i.e. does not contain impurities), so that no blowdown apparatus is required to remove impurities. Further, as heating apparatuses typically use a vapour supply in a heating operation, such a cooling unit may be easily assembled with a heating apparatus by providing the separate cooling unit with a heat exchanger and connecting it to use the same vapour supply as the heating apparatus. Therefore, no separate condensate or liquid supply would be required.

(20) The cooling unit **50** further comprises a venting line **60** which is connected to the vapour line **18** between the supply valve **22** and the heating chamber **12**, so that it is fluidically connected to the cavity of the heating chamber **12**. The venting line **60** comprises a vent valve **62** which is configured to permit venting of the heating chamber **12** to atmosphere when the vent valve **62** is in an open position.

(21) The venting line **60** is vented to atmosphere through a pressure reducer **64**. In this example, the pressure reducer **64** is a thermocompressor, with the venting line **60** is connected to a vapour inlet of the thermocompressor **64**. A motive line **66** fluidically connects the steam supply **20** with a motive inlet of the thermocompressor **64**. The motive line **66** comprises a motive valve **68** which is configured to permit through-flow of motive steam from the steam supply **20** to the motive inlet of the thermocompressor **64** when the motive valve **68** is in an open position.

(22) The thermocompressor **64** is thereby configured to pump vapour from the heating chamber **14** to atmosphere when both the vent valve **62** and the motive valve **68** are in open positions, and to

allow venting of steam (or any vapour) from the heating chamber **14** to atmosphere when the vent valve **62** is open, and the motive valve **68** is closed. The term venting as used herein is intended to mean passively permitting a pressurised vapour to exit the chamber under its own motive force, whereas pumping is intended to mean actively driving a vapour from the heating chamber, for example by using a pressure reducer such as the thermocompressor. In some examples, the pressure reducer may be any device capable of reducing a downstream pressure to draw vapour from the heating chamber, such as a vacuum pump. In other examples, there may be no pressure reducer, such that the vent valve merely permits venting of the heating chamber to ambient air.

(23) The cooling unit **50** also comprises a controller **70** which is configured to receive the pressure parameter and temperature parameter from the pressure sensor **102** and temperature sensor **104** respectively. The controller **70** is configured to control the opening and closing of the condensate supply valve **56**, the vent valve **62** and the motive valve **68** based on the received temperature and pressure parameters to keep a cooling rate of the heating chamber **14** within an acceptable limit, as will be explained in more detail below.

(24) In this example, the cooling unit **50** is provided on a skid or pallet, such that it can be easily retrofitted as a module to the heating apparatus, and equally easily removed from the heating apparatus **10**. Providing the cooling unit **50** on a skid or pallet also allows the cooling unit **50** to be easily moved between different heating apparatuses, so that it can be retrofitted to a heating apparatus, removed from that heating apparatus after the cooling operation has been performed, and moved to a different heating apparatus.

(25) In this example, as explained above, the supply valve **22** and the condensate valve **28** are open during the heating operation. At the end of the heating operation, a cooling operation may be commenced which will be described below.

(26) FIG. **3** is a flow chart **200** showing steps of a cooling operation in which a heating chamber **14** as described above is cooled in a heating installation **100**. The cooling operation is initiated after a heating operation in which the heating chamber **14** has been heated.

(27) In block **202**, the heating chamber **14** is isolated from the steam supply **20** and from the condensate drain **26**. The heating chamber **14** is isolated from the steam supply **20** by closing the supply valve **22**, and the heating chamber **14** is isolated from the condensate drain **26** by closing the condensate valve **28**. In other examples where the heating apparatus does not comprise a vapour supply and is heated by other means, this step is not performed. In other examples where the heating apparatus alternatively or additionally does not include a condensate drain, this step of the method may not include isolating the heating chamber from the condensate drain.

(28) In block **204**, condensate is accumulated in the heating chamber **14** from the condensate supply **52**. Condensate is accumulated by opening the condensate supply valve **56** such that steam from the steam supply **20** is permitted to flow through the heat exchanger **54**. The steam is condensed to condensate in the heat exchanger **54**, and the condensate is permitted to flow through the steam trap **58**, into the heating chamber **12**. In other examples where the cooling unit does not include a heat exchanger, a corresponding step may instead include opening a condensate supply valve to allow flow of condensate from any condensate supply, or may include activating a pump to pump condensate into the heating chamber. When a predetermined amount of condensate has been accumulated in the heating chamber **14**, the controller **70** closes the condensate supply valve **56**. The predetermined amount of condensate may be calculated specifically to cool the chamber by a predetermined amount. In this example no more condensate from the condensate supply **52** is permitted to accumulate in the heating chamber **12** during the cooling operation.

(29) In some examples, there is no controller such that the opening and closing of the condensate supply valve **56** may be controlled and performed manually. The amount of condensate accumulated may be determined with a flow sensor, or any other suitable sensor.

(30) In block **206**, the pressure in the heating chamber **12** is monitored by the controller **70**. A pressure parameter from a pressure sensor **102** and a temperature parameter from a temperature

sensor **104** are received by the controller **70** and the controller **70** monitors the pressure and temperature based on the received parameters. The pressure and temperature monitoring also occurs continuously during block **208**. In some examples, the pressure and temperature may be displayed to a user, and may thereby be monitored by a user instead of or as well as by a controller.

(31) In block **208**, the controller **70** controls the pressure in the heating chamber and thereby the cooling of the heating chamber **14** by controlling venting of the heating chamber **12**. In some examples, the venting of the heating chamber may be manually controlled by a user based on displayed pressure and temperature parameters from the pressure and temperature sensors.

(32) FIG. **4** shows an example sub-method **208** of controlling venting of the heating chamber **14**. During the sub-method **208**, the pressure and temperature in the heating chamber **14** is continuously monitored by the controller **70**.

(33) In block **252**, the heating chamber **14** is driven to rotate about the longitudinal axis **15**. The heating chamber **14** is rotated throughout the sub-method **208** to even out the cooling of the heating chamber **14** across the whole surface of the drum **14** during cooling. For example, a rate of heat transfer may be higher where condensate is in contact with a wall of the chamber **14** than where vapour is in contact with the wall of the chamber **14**. In other examples, the heating chamber **14** may remain stationary during the cooling operation, or may begin rotating earlier in the cooling operation (for example during or before accumulation of condensate).

(34) In block **254** of the sub-method **208**, the controller **70** initiates and controls a first phase of venting, in which the vent valve **62** is opened to permit venting of the heating chamber **14** to atmosphere. When the pressure in the heating chamber is above atmospheric pressure, opening of the vent valve **62** results in a pressure drop in the heating chamber **12** such that the saturation temperature reduces and the condensate vapourises. The applicant has found that this promotes heat transfer from the walls of the heating chamber **14** to the fluid received therein, both owing to a temperature differential between the wall and the fluid as the saturation temperature reduces, and owing to heat transfer corresponding to the latent heat of vaporisation of the condensate as it vaporises. The vaporised condensate (steam) is vented through the vent valve **62**. By controlling the rate of reduction in pressure in the heating chamber **14** the method enables more controlled and faster heat transfer from the heating chamber **14** than is possible by passive cooling. In an example, the applicant has found that a cooling time of a heating chamber may be reduced from 8 hours to 2 hours.

(35) The controller **70** monitors the pressure in the heating chamber **14** and the temperature of the heating chamber **14** during the first phase of venting, and controls the opening and closing of the vent valve **62** to control the reduction in pressure in the heating chamber **14** to thereby maintain a rate of cooling of the heating chamber **14** within a predetermined range and/or to control the cooling of the heating apparatus to follow a predetermined cooling profile.

(36) In block **256**, the controller **70** determines whether the pressure within the heating chamber has reached a pressure threshold corresponding to initiating pumping. In this example, the pressure threshold is atmospheric pressure (approximately 101 kPa at sea level), as the vapour is being vented to atmospheric conditions. When the pressure within the heating chamber **14** reaches the pressure threshold (i.e. 101 kPa in this example), passive venting would stop such that vaporisation of condensate in the heating chamber **14** also stops. Therefore, further cooling of the condensate and vapour within the heating chamber below saturation temperature (100° C. at atmospheric conditions) would only be via heat transfer from the exterior of the drum.

(37) In the example method, when it is determined that the pressure in the heating chamber **14** has reached the pressure threshold, the controller **70** moves on to block **258** to cause further reduction of pressure.

(38) In block **258**, the controller **70** initiates a second phase of venting. In the second phase of venting, the controller **70** activates the pressure reducer **64**, which in this example involves opening the motive valve **68** to activate the thermocompressor **64**. When the vent valve **62** is opened, the

thermoc compressor **64** pumps the steam out of the heating chamber **14**, thereby reducing the pressure in the heating chamber **12** to below the pressure threshold. This causes further reduction of the saturation temperature and further vaporisation of the condensate, thereby further cooling the heating chamber **14**. The controller **70** continues to monitor the pressure and temperature and to control opening and closing of the vent valve **62** while the motive valve **68** is open, so as to maintain the rate of cooling of the heating chamber **14** to within the predetermined range until the temperature of the heating chamber **12** and/or drum **14** reaches the desired cooled temperature, which in this example is approximately 50° C. This is the end of the cooling operation, such that the heating chamber may be cleaned, and the heating operation and cooling operation may be repeated.

(39) An example predetermined range for a rate of cooling of the heating chamber may be between 0.5-10° C. per minute, such as 0.5-5° C. per minute, or 0.5-3° C. per minute or 0.5-1.5° C. per minute. Different rates of cooling may be suitable for different chambers and materials.

(40) As described above, in some examples, there may be no pressure reducer in the cooling unit, such that there is no second phase of venting as described above. In such examples, cooling is controlled in a first phase of venting only.

(41) In this example, the amount of condensate accumulated in the heating chamber **12** at the beginning of the method **200** may be calculated such that it is substantially all vaporised at the point when the temperature of the heating chamber reaches the desired cooled temperature.

(42) Although it has been described that a predetermined amount of condensate is accumulated in the heating chamber before venting is initiated in the second step **204** of the method **200**, and that no further condensate is permitted to accumulate in the heating chamber, in other examples, the accumulation of condensate in the heating chamber may occur simultaneously or alternately with venting of the heating chamber. In such examples, the time taken to cool the heating chamber to the desired cooled temperature may be further reduced because the cooling of the heating chamber by venting may be initiated immediately on initiation of the cooling operation, rather than waiting for the predetermined amount of condensate to be accumulated, and only then initiating the cooling by permitting venting of the heating chamber.

(43) Although it has been described that a one heating chamber receives vapour from a vapour supply, in some examples, there may be multiple heating chambers which receive vapour from a common vapour supply.

(44) Although it has been described that a controller receives the pressure parameter and the temperature parameter and controls the opening and closing of the condensate supply valve, vent valve, and motive valve, in other examples, there may be no controller for one or more of these operations. In such examples, the pressure parameter and temperature parameter may be displayed to a user, and the user may control the opening and closing of the valves using the method as described.

(45) Whilst an example cooling operation has been described in which venting is controlled to maintain a rate of cooling of the heating chamber within a predetermined range, it should be appreciated that this may be achieved either by monitoring a parameter relating to the temperature of the heating chamber and controlling the venting in response, or by causing the venting to occur in a predetermined way that is known to correspond to a cooling of the heating chamber at a rate within the predetermined range. This may be determined empirically, for example during commissioning of a heating installation, or by simulation.

(46) In other examples, the heating chamber may take any suitable form, and may be configured to heat an article on the outside of the heating chamber, or to heat articles on the interior of the heating chamber. In some examples, articles for heating may be received in the cavity of the heating chamber (e.g. a steam oven). In other examples, the heating chamber may be in the form of a jacket having an internal and external wall with the cavity defined therebetween, and an article

for heating disposed on the opposite side of the internal wall from the cavity (i.e. internally within a separate nested cavity).

Claims

1. A method of heating and cooling a heating apparatus comprising a heating chamber, the method comprising: accumulating condensate in the heating chamber, wherein accumulating the condensate in the heating chamber comprises receiving vapour from a vapour supply into the heating apparatus to heat the heating chamber and condensing the vapour in the heating chamber; monitoring a pressure in the heating chamber; controlling the pressure in the heating chamber by venting the heating chamber to thereby controllably permit vaporisation of the condensate that cools the heating chamber; and draining condensate from the heating chamber via a condensate line; wherein the pressure is controlled to maintain a rate of cooling of the heating chamber within a predetermined range.
2. A method according to claim 1, wherein vapour is vented in a first phase until the pressure in the heating chamber is reduced to a threshold pressure; and wherein the method comprises activating a pressure reducer in response to reaching the threshold pressure, to reduce the pressure in the heating chamber below the threshold pressure in a second phase, in order to further control the vaporisation of the condensate.
3. A method according to claim 1, wherein the heating apparatus is coupled to a vapour supply, and wherein the method comprises isolating the vapour supply from the heating chamber before accumulating the condensate.
4. A method according to claim 1, wherein the heating apparatus is coupled to a condensate drain via the condensate line, and wherein the method comprises isolating the heating chamber from the condensate drain before accumulating the condensate.
5. A method of cooling a heating apparatus comprising a heating chamber, the method comprising: accumulating condensate in the heating chamber; monitoring a pressure in the heating chamber; controlling the pressure in the heating chamber by venting the heating chamber in a first phase to thereby controllably permit vaporisation of the condensate that cools the heating chamber; and activating a pressure reducer in response to reaching a threshold pressure, to reduce the pressure in the heating chamber below the threshold pressure in a second phase, in order to further control the vaporisation of the condensate; wherein the pressure is controlled to maintain a rate of cooling of the heating chamber within a predetermined range, wherein the heating chamber is vented through a vent valve, and wherein the pressure reducer is a thermocompressor which is activated by opening a motive valve connecting a vapour supply to a motive inlet of the thermocompressor, and wherein the heating chamber is connected to a vapour inlet of the thermocompressor via the vent valve.
6. A method according to claim 1, wherein a predetermined amount of condensate is accumulated in the heating chamber before venting of the heating chamber, and wherein no further condensate is accumulated after the venting begins until cooling operation is completed.
7. A method according to claim 1, wherein the pressure of the heating chamber is automatically monitored by a controller, and the venting of the heating chamber is automatically controlled with the controller based on the monitored pressure, so that cooling of the heating apparatus follows a predetermined cooling profile.
8. A method according to claim 1, comprising monitoring a rate of cooling, and controlling the pressure based on the monitored rate of cooling, such that the rate of cooling stays within a predetermined range.
9. A heating installation configured to perform the method in accordance with claim 1, the heating installation comprising: a heating chamber; a vapour supply configured to supply vapour to the heating chamber to heat the heating chamber and condense in the heating chamber and to form a

condensate; a vent valve, and a venting line connecting the heating chamber to the vent valve, which are configured to permit venting of the heating chamber to evaporate the condensate; a pressure sensor configured to generate a pressure parameter relating to the pressure in the heating chamber; a condensate drain coupled to the heating chamber via a condensate line for draining condensate from the heating chamber in a heating operation, and wherein the condensate line comprises a condensate valve configured to isolate the condensate drain from the heating chamber; and a controller configured to control the heating installation to perform: accumulating condensate in the heating chamber; monitoring the pressure in the heating chamber; controlling the pressure in the heating chamber by venting the heating chamber via the vent valve and the venting line, to thereby controllably permit vaporisation of the condensate that cools the heating chamber; and draining condensate from the heating chamber via the condensate line, wherein the pressure is controlled to maintain a rate of cooling of the heating chamber within a predetermined range, and wherein accumulating condensate in the heating chamber comprises receiving the vapour from the vapour supply into the heating apparatus to heat the heating chamber and condensing the vapour in the heating chamber.

10. A heating installation according to claim 9, further comprising a pressure reducer configured to reduce pressure in the heating chamber below a threshold pressure when activated.

11. A heating installation according to claim 9, further comprising a condensate supply comprising the vapour supply and a heat exchanger configured to condense the vapour from the vapour supply to produce the condensate.

12. A heating installation according to claim 11, comprising a vapour line connecting the vapour supply with the heating chamber for supplying the vapour to heat the heating chamber in a heating operation, and wherein the vapour line comprises a supply valve configured to isolate the vapour supply from the heating chamber.

13. A cooling unit for connecting to a heating apparatus having a heating chamber, for cooling the heating apparatus using a method in accordance with claim 1, the cooling unit comprising: a venting line configured to be connected to the heating chamber, the venting line having a vent valve for permitting venting of the heating chamber; a condensate line configured to be connected to the heating chamber and having a condenser configured to supply condensate to the heating chamber via the condensate line, and wherein the condensate line comprises a condensate drain between an outlet of the condenser and the heating chamber.

14. A cooling unit according to claim 13, comprising a pressure reducer connected to the vent valve via the venting line and configured to reduce a pressure in the heating chamber, and optionally wherein the pressure reducer comprises a thermocompressor having a motive inlet which is connected to a vapour line, the vapour line having a motive valve and being connectable to a vapour supply for supplying motive vapour to the thermocompressor.

15. A cooling unit according to claim 13, comprising a vapour generator for supplying vapour to an inlet of the condenser, which is configured to condense the vapour to supply condensate.

16. A cooling unit according to claim 13, wherein the cooling unit is configured to be connectable to and detachable from a vapour supply which is configured to supply vapour to the heating apparatus, so as to supply vapour to the condenser for supplying condensate when connected.

17. A cooling unit according to claim 13, wherein the condensate line of the cooling unit is configured to be connectable to a condensate line of the heating apparatus.

18. A cooling unit according to claim 13, comprising a sensor configured to generate a pressure parameter relating to a pressure in the heating chamber, and/or comprising a controller configured to monitor the pressure in the heating chamber and to automatically control opening of the vent valve so that the cooling of the heating apparatus follows a predetermined cooling profile and/or activation of the pressure reducer so that the cooling of the heating apparatus follows a predetermined cooling profile.

19. A method of heating and cooling a heating apparatus comprising a heating chamber, the method

comprising: accumulating condensate in the heating chamber; monitoring a pressure in the heating chamber; controlling the pressure in the heating chamber by venting the heating chamber to thereby controllably permit vaporisation of the condensate that cools the heating chamber; and draining condensate from the heating chamber via a condensate line; wherein the pressure is controlled to maintain a rate of cooling of the heating chamber within a predetermined range; wherein accumulating the condensate comprises receiving vapour from a vapour supply, condensing the vapour in a heat exchanger and supplying the condensate to the heating chamber; and wherein the vapour supply is the same vapour supply as is used in a previous heating operation to heat the heating apparatus.
